When $H = \boldsymbol{p}^T \boldsymbol{M}^{-1}\boldsymbol{p}/2 + U(\boldsymbol{q})$ this is just the leapfrog/Verlet method, but it can be used also for more general systems. In the more general setting it is implicit (a nonlinear set of equations must be solved to advance from time level to time level). To see that it is symplectic, we first note that this is a symmetric composition of the form

$$\mathcal{K}_h = \mathcal{G}_{h/2}^{\dagger} \circ \mathcal{G}_{h/2},$$

where $\mathcal{G}_h$ is defined by

$$\boldsymbol{P} = \boldsymbol{p} - h\nabla_{\boldsymbol{q}} H(\boldsymbol{q}, \boldsymbol{P}), \tag{2.29}$$

$$\boldsymbol{Q} = \boldsymbol{q} + h\nabla_{\boldsymbol{p}} H(\boldsymbol{q}, \boldsymbol{P}), \tag{2.30}$$

so it is enough to show that this basic method is symplectic. Taking differentials of (2.30) defining $\mathcal{G}_h$ and then wedge products and summing, we have

$$\sum_i \mathrm{d}Q_i \wedge \mathrm{d}P_i = \sum_i \mathrm{d}q_i \wedge \mathrm{d}P_i + h \sum_i \sum_j H_{p_i q_j} \mathrm{d}q_j \wedge \mathrm{d}P_i$$
$$+ h \sum_i \sum_j H_{p_i p_j} \mathrm{d}P_j \wedge \mathrm{d}P_i.$$

The last term on the right vanishes by equality of mixed partials and the antisymmetry of the wedge product. On the other hand, using (2.29), we obtain, by similar means,

$$\sum_i \mathrm{d}q_i \wedge \mathrm{d}P_i = \sum_i \mathrm{d}q_i \wedge \mathrm{d}p_i - h \sum_i \sum_j H_{q_i p_j} \mathrm{d}q_i \wedge \mathrm{d}P_j.$$

Relabelling the indices in this sum and using our previous work results in

$$\sum_i \mathrm{d}Q_i \wedge \mathrm{d}P_i = \sum_i \mathrm{d}q_i \wedge \mathrm{d}p_i,$$

implying that the method is symplectic.

The more general family of Partitioned Runge-Kutta methods is defined by making use of a partitioning of the system and introducing combinations of a set of internal stages. This more general family of schemes is discussed in some detail in [326] (see also discussions of [164, 227]).